

Lecture 5 Tutorial — Answers

Q1. Bayes' Rule Components

1. **A – 1** $P(c)$: Prior probability
2. **B – 2** $P(x | c)$: Likelihood
3. **C – 3** $P(c | x)$: Posterior probability
4. **D – 4** $P(x)$: Evidence factor (observation)

Q2. Likelihood Computation (Coin Models)

Given

$$D = \text{THHTTTHTTH}$$

we count:

$$H = 4, \quad T = 6$$

1. Number of H is **4**, number of T is **6**.

2.

$$L_1 = p(D | M_1) = (0.5)^4(0.5)^6 = (0.5)^{10} = \frac{1}{1024}$$

3.

$$L_2 = p(D | M_2) = (0.4)^4(0.6)^6$$

4. Since

$$(0.4)^4(0.6)^6 > (0.5)^{10},$$

the model with higher likelihood is M_2 .

Q3. Log-likelihood (Numerical Stability)

1.

$$\begin{aligned}\Delta &= \log p(D | M_2) - \log p(D | M_1) \\ \Delta &= 4 \log(0.4) + 6 \log(0.6) - 10 \log(0.5)\end{aligned}$$

2. Since $L_2 > L_1$, we have

$$\Delta > 0$$

Q4. Maximum Likelihood Estimation (MLE) for a Coin

For Bernoulli trials with $\theta = p(H)$:

1.

$$p(D | \theta) = \theta^4(1 - \theta)^6$$

2. The MLE is

$$\hat{\theta} = \frac{4}{10} = 0.4$$

Q5. Occam's Razor / Size Principle

Let

$$D = \{16, 8, 2, 64\}$$

1. Powers of two under 100 are

$$\{1, 2, 4, 8, 16, 32, 64\}$$

so

$$|h_{\text{two}}| = 7$$

2. Even numbers below 100 are

$$2, 4, 6, \dots, 98$$

so

$$|h_{\text{even}}| = 49$$

3. Under the uniform assumption,

$$p(D \mid h_{\text{two}}) = \left(\frac{1}{7}\right)^4, \quad p(D \mid h_{\text{even}}) = \left(\frac{1}{49}\right)^4$$

hence

$$\frac{p(D \mid h_{\text{two}})}{p(D \mid h_{\text{even}})} = \frac{(1/7)^4}{(1/49)^4} = \left(\frac{49}{7}\right)^4 = 7^4 = 2401$$

Q6. Posterior Update (Equal Priors)

Assume

$$p(h_{\text{two}}) = p(h_{\text{even}}) = 0.5$$

and a new sample $x_{\text{new}} = 32$ arrives.

1. Since 32 is consistent with both hypotheses,

$$p(D \cup \{32\} \mid h_{\text{two}}) = \left(\frac{1}{7}\right)^5, \quad p(D \cup \{32\} \mid h_{\text{even}}) = \left(\frac{1}{49}\right)^5$$

Therefore the unnormalised posteriors are

$$p(h_{\text{two}} \mid D \cup \{32\}) \propto 0.5 \left(\frac{1}{7}\right)^5$$

$$p(h_{\text{even}} \mid D \cup \{32\}) \propto 0.5 \left(\frac{1}{49}\right)^5$$

2. The larger posterior is h_{two} .

Q7. Entropy, Cross Entropy, KL Divergence

Given

$$P = (0.5, 0.5), \quad Q = (0.9, 0.1)$$

1. Using natural logarithm,

$$\begin{aligned} KL(P \parallel Q) &= 0.5 \ln \frac{0.5}{0.9} + 0.5 \ln \frac{0.5}{0.1} \\ &= 0.5 \ln \left(\frac{5}{9}\right) + 0.5 \ln(5) = 0.5 \ln \left(\frac{25}{9}\right) = \ln \left(\frac{5}{3}\right) \end{aligned}$$

So

$$KL(P \parallel Q) = \ln \left(\frac{5}{3}\right)$$

2. **No**, $KL(P \parallel Q)$ cannot be negative.

Q8. Zero-point Problem and Laplacian Correction

Given counts

$$N_a = 2, \quad N_b = 0, \quad N_c = 1$$

the total count is

$$N = 3$$

1. Unsmoothed MLE probabilities:

$$p(a) = \frac{2}{3}, \quad p(b) = 0, \quad p(c) = \frac{1}{3}$$

2. With Laplacian correction (add one to each count):

$$N'_a = 3, \quad N'_b = 1, \quad N'_c = 2$$

new total:

$$3 + 1 + 2 = 6$$

so the smoothed probabilities are

$$p(a) = \frac{3}{6} = \frac{1}{2}, \quad p(b) = \frac{1}{6}, \quad p(c) = \frac{2}{6} = \frac{1}{3}$$

Q9. MAP vs MLE

C. MAP maximises $p(c | x)$ and uses a term proportional to $p(c)p(x | c)$.

Q10. Naive Bayes Independence Assumption

The features are **independent** given the class c .

Q11. Naive Bayes Classification

The dataset has 14 samples:

$$9 \text{ Yes}, \quad 5 \text{ No}$$

so

$$p(\text{Yes}) = \frac{9}{14}, \quad p(\text{No}) = \frac{5}{14}$$

The query is

$$x = (\text{Outlook}=\text{Overcast}, \text{Temperature}=\text{Cool}, \text{Humidity}=\text{Normal}, \text{Windy}=\text{True})$$

Part A — No smoothing

For class Yes:

$$p(\text{Overcast} | \text{Yes}) = \frac{4}{9}, \quad p(\text{Cool} | \text{Yes}) = \frac{3}{9}, \quad p(\text{Normal} | \text{Yes}) = \frac{6}{9}, \quad p(\text{True} | \text{Yes}) = \frac{3}{9}$$

Thus

$$S_{\text{Yes}} = \frac{9}{14} \cdot \frac{4}{9} \cdot \frac{3}{9} \cdot \frac{6}{9} \cdot \frac{3}{9} = \frac{4}{189}$$

For class No:

$$p(\text{Overcast} | \text{No}) = \frac{0}{5} = 0$$

so

$$S_{\text{No}} = 0$$

- 1.

$$S_{\text{Yes}} = \frac{4}{189}, \quad S_{\text{No}} = 0$$

2. Predicted class: **Yes**

Part B — With Laplacian correction

Category counts:

$$|\text{Outlook}| = 3, \quad |\text{Temperature}| = 3, \quad |\text{Humidity}| = 2, \quad |\text{Windy}| = 2$$

For class Yes ($n_{\text{Yes}} = 9$):

$$p(\text{Overcast} \mid \text{Yes}) = \frac{4+1}{9+3} = \frac{5}{12}$$

$$p(\text{Cool} \mid \text{Yes}) = \frac{3+1}{9+3} = \frac{4}{12} = \frac{1}{3}$$

$$p(\text{Normal} \mid \text{Yes}) = \frac{6+1}{9+2} = \frac{7}{11}$$

$$p(\text{True} \mid \text{Yes}) = \frac{3+1}{9+2} = \frac{4}{11}$$

Hence

$$S_{\text{Yes}} = \frac{9}{14} \cdot \frac{5}{12} \cdot \frac{1}{3} \cdot \frac{7}{11} \cdot \frac{4}{11} = \frac{5}{242}$$

For class No ($n_{\text{No}} = 5$):

$$p(\text{Overcast} \mid \text{No}) = \frac{0+1}{5+3} = \frac{1}{8}$$

$$p(\text{Cool} \mid \text{No}) = \frac{1+1}{5+3} = \frac{2}{8} = \frac{1}{4}$$

$$p(\text{Normal} \mid \text{No}) = \frac{1+1}{5+2} = \frac{2}{7}$$

$$p(\text{True} \mid \text{No}) = \frac{3+1}{5+2} = \frac{4}{7}$$

Thus

$$S_{\text{No}} = \frac{5}{14} \cdot \frac{1}{8} \cdot \frac{1}{4} \cdot \frac{2}{7} \cdot \frac{4}{7} = \frac{5}{2744}$$

1.

$$S_{\text{Yes}} = \frac{5}{242}, \quad S_{\text{No}} = \frac{5}{2744}$$

2. Predicted class: **Yes**

Q12. Bayesian Network

B. It encodes feature dependencies via a DAG and expresses a joint distribution as a product of conditional distributions.